SHARDA UNIVERSITY

Beyond Boundaries

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Section- CS I

Group 1

Working of Code

Large Language Models (LLMs) cannot efficiently process very long text at once. So we **split text into smaller chunks** before:

- Sending it to ChatGPT / LLMs
- Creating embeddings
- Storing in vector databases

This notebook explains **5 levels of text splitting**, from **basic to advanced**.

Level 1: Character Splitting

Code:

python

Copy code

text = "This is the text I would like to chunk up..."

python

Copy code

chunks = [] chunk_size = 35 for i in range(0, len(text), chunk_size): chunk = text[i:i + chunk_size] chunks.append(chunk)

Working:

- The text is split **purely by character count**
- Every chunk has **35 characters**
- Meaning and sentence structure are ignored

Output Example:

css

Copy code

"This is the text I would like "

"to chunk up. It is the example"

"text for this exercise"

❌ **Drawback:**

- Sentences may break mid-word
- No semantic meaning preserved

Level 2: Sentence Splitting

Working:

- Text is split using **sentence boundaries**
- Usually using . or NLP libraries

Why better?

- Keeps complete sentences
- Improves LLM understanding

Example:

nginx

Copy code

Sentence 1 → Chunk 1

Sentence 2 → Chunk 2

Level 3: Paragraph Splitting

Working:

- Splits text by **paragraphs**
- Uses newline characters (\n\n)

Benefit:

- Preserves context better than sentence splitting
- Ideal for articles, blogs, documentation

Level 4: Token-Based Splitting

Working:

- Splits text based on **token count**, not characters
- Tokens ≠ words (used by models like GPT)

Why important?

- LLMs have **token limits**
- Prevents overflow errors

Used in:

- OpenAI
- Hugging Face
- LangChain

Level 5: Recursive / Semantic Splitting (Best)

Working:

- Tries multiple splitting strategies:
    1. Paragraphs
    2. Sentences
    3. Words
    4. Characters (last fallback)

Goal:

- Keep chunks **under a size limit**
- Preserve **maximum semantic meaning**

Used in:

- RAG (Retrieval-Augmented Generation)
- Chatbots with PDFs
- Document QA systems